

Validation du sentiment BERT avec les notes Airbnb

VALIDATION DU SENTIMENT BERT AVEC LES NOTES AIRBNB

Nombre de lignes tabulaire : 53745

Nombre d'annonces avec sentiment BERT : 41923

Nombre d'annonces exploitables avec review_scores_rating et BERT : 41923

Corrélations principales avec review_scores_rating :

- bert_stars_moyen : Pearson=0.8446, Spearman=0.7168
- sentiment_bert_moyen : Pearson=0.8446, Spearman=0.7168
- ratio_avis_bert_positifs : Pearson=0.7924, Spearman=0.6668
- sentiment_bert_min : Pearson=0.3786, Spearman=0.5881
- bert_confiance_moyenne : Pearson=0.3238, Spearman=0.4605
- sentiment_bert_max : Pearson=0.6588, Spearman=0.1706
- nb_avis_bert_positifs : Pearson=0.1023, Spearman=-0.0967
- nb_avis_textuels_bert : Pearson=0.0899, Spearman=-0.1513
- nb_avis_bert_neutres : Pearson=-0.0705, Spearman=-0.4024
- ratio_avis_bert_neutres : Pearson=-0.3122, Spearman=-0.4616

Validation directe bert_stars_moyen vs review_scores_rating :

- MAE : 0.1913
- RMSE : 0.3184
- R2 : 0.6073

Régression train/test : variables BERT -> review_scores_rating

- MAE test : 0.1463
- RMSE test : 0.2553
- R2 test : 0.7257

Classification directe classes BERT vs classes note Airbnb

- Accuracy : 0.8200
- Balanced accuracy : 0.7921
- F1 macro : 0.7053

Classification train/test : variables BERT -> classe note Airbnb

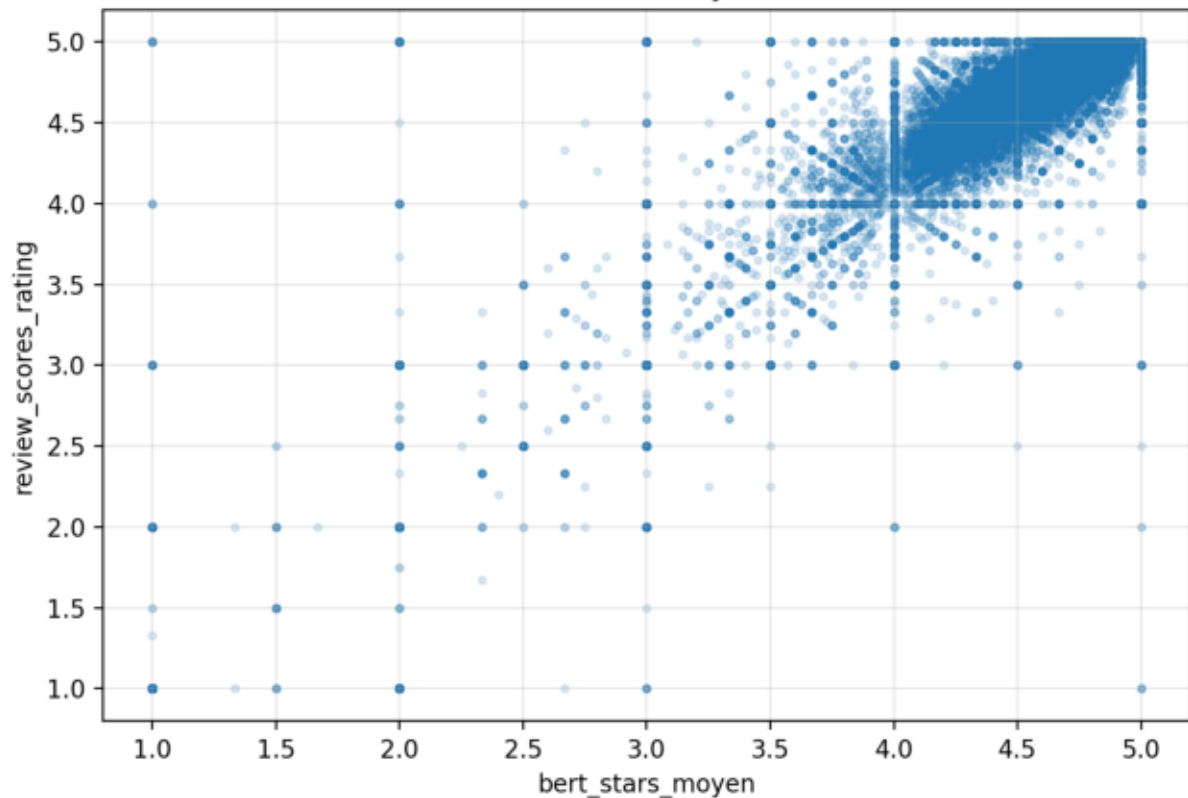
- Accuracy : 0.8140
- Balanced accuracy : 0.7860
- F1 macro : 0.7034
- ROC-AUC macro : 0.9269

Validation binaire : détecter review_scores_rating < 4.5

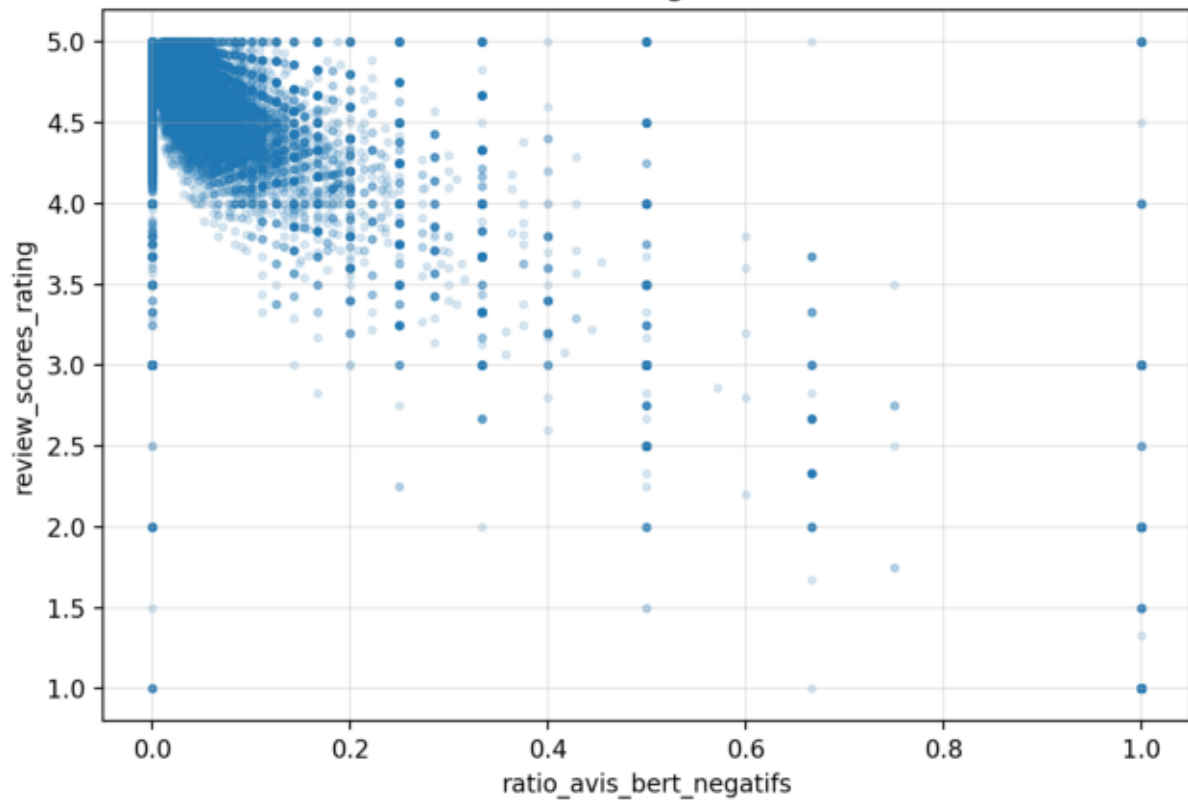
Corrélations complètes

variable_bert	n	pearson_correlation	pearson_pvalue	spearman_correlation	spearman_pvalue
bert_stars_moyen 41923		0.844603	0.000000e+00	0.716790	0.000000e+00
sentiment_bert_moyen 41923		0.844603	0.000000e+00	0.716790	0.000000e+00
ratio_avis_bert_positifs 41923		0.792373	0.000000e+00	0.666840	0.000000e+00
sentiment_bert_min 41923		0.378623	0.000000e+00	0.588124	0.000000e+00
bert_confiance_moyenne 41923		0.323807	0.000000e+00	0.460466	0.000000e+00
sentiment_bert_max 41923		0.658836	0.000000e+00	0.170640	2.292660e-271
nb_avis_bert_positifs 41923		0.102305	6.344217e-98	-0.096706	1.176977e-87
nb_avis_textuels_bert 41923		0.089939	5.037752e-76	-0.151336	3.179835e-213
nb_avis_bert_neutres 41923		-0.070452	2.786119e-47	-0.402385	0.000000e+00
ratio_avis_bert_neutres 41923		-0.312162	0.000000e+00	-0.461571	0.000000e+00
nb_avis_bert_negatifs 41923		-0.153481	2.535010e-219	-0.463050	0.000000e+00
ratio_avis_bert_negatifs 41923		-0.767638	0.000000e+00	-0.529573	0.000000e+00
sentiment_bert_ecart_type 41923		-0.304337	0.000000e+00	-0.621103	0.000000e+00

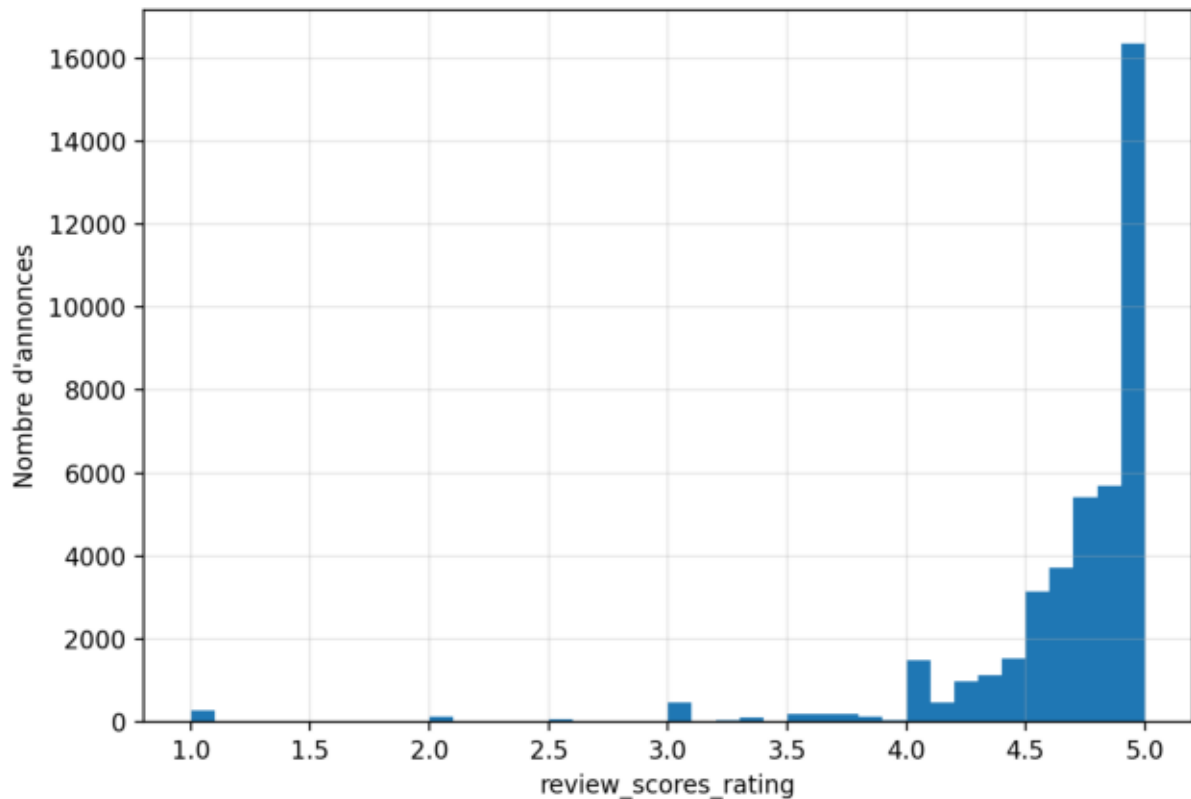
Relation entre étoiles BERT moyennes et note Airbnb



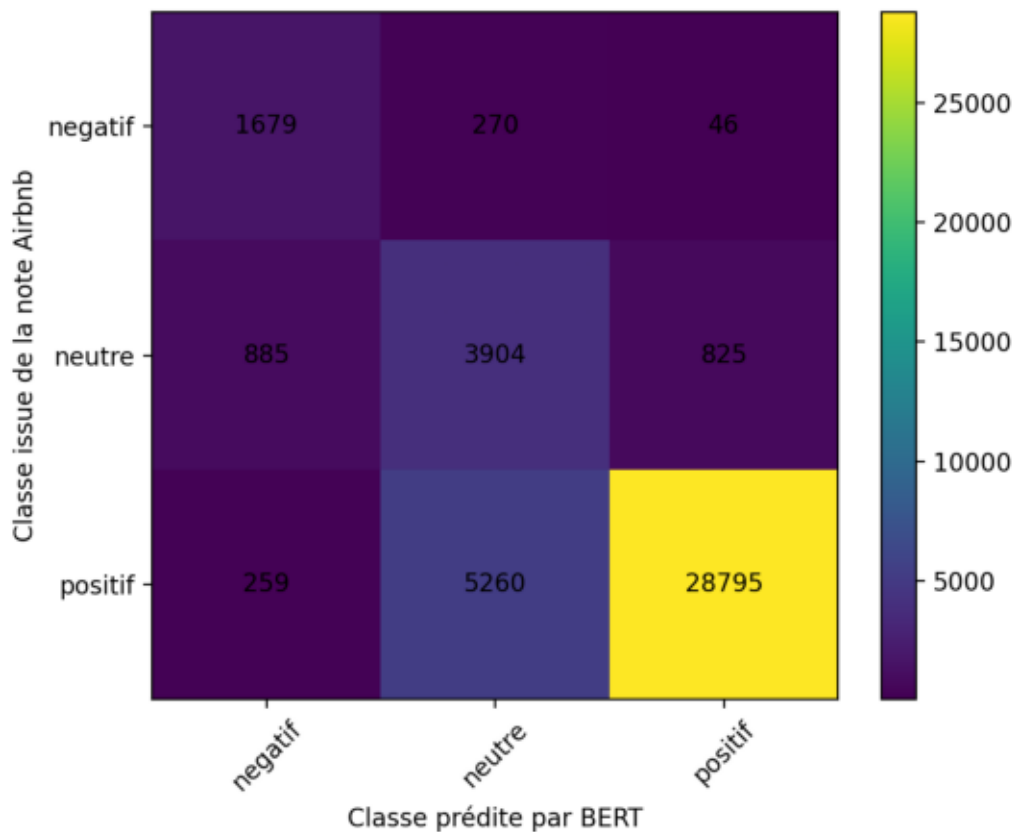
Relation entre ratio d'avis négatifs BERT et note Airbnb



Distribution des notes Airbnb



Matrice de confusion : classes note Airbnb vs classes BERT



Rapport classification direct

	precision	recall	f1-score	support
negatif	0.59	0.84	0.70	1995
neutre	0.41	0.70	0.52	5614
positif	0.97	0.84	0.90	34314
accuracy			0.82	41923
macro avg	0.66	0.79	0.71	41923
weighted avg	0.88	0.82	0.84	41923

Rapport classification train/test

	precision	recall	f1-score	support
negatif	0.61	0.82	0.70	399
neutre	0.40	0.70	0.51	1123
positif	0.97	0.83	0.89	6863
accuracy			0.81	8385
macro avg	0.66	0.79	0.70	8385
weighted avg	0.88	0.81	0.83	8385